

❏ What is IIS?

IIS (Internet Information Services) is a **flexible, secure and manageable** web server developed by **Microsoft**. It runs on **Windows Server** and some client versions of Windows. IIS is used to **host and serve websites and web applications**, supporting technologies like **HTML, ASP.NET, PHP**, and more.

❏ Core Functions of IIS

Function	Description
HTTP Web Hosting	Hosts websites and serves web pages to users via the HTTP/HTTPS protocol.
Application Hosting	Hosts web apps developed in ASP.NET, PHP, or other languages.
Security Management	Supports SSL certificates, authentication, and authorization settings.
Logging & Monitoring	Tracks server activity via logs and performance counters.
Load Balancing	Distributes incoming requests across multiple servers (via ARR or Network Load Balancing).

☒ IIS Architecture Overview

1.
Web Server Engine: Handles incoming requests and responses.
 2.
Worker Process (w3wp.exe): Runs web applications and processes requests.
 3.
Application Pool: Isolates applications for security and reliability.
 4.
Modules: Plug-ins that handle specific tasks (e.g., authentication, compression, caching).
 5.
Configuration Files: applicationHost.config and web.config store settings.
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☒ Supported Protocols

- HTTP / HTTPS

- FTP / FTPS
 - SMTP (optional)
 - WebSocket
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☒ Key Components

Component	Purpose
Application Pools	Provides process isolation between applications.
Sites	Logical containers for hosting web applications.
Bindings	Assign IP + port + hostname to a site (e.g., 80, 443, etc.).
Virtual Directories	Points to content outside the site's root folder.
Modules & Handlers	Define how IIS handles requests (e.g., authentication, errors).

☒ Security Features

- **Authentication:** Basic, Digest, Windows, Forms.
 - **Authorization Rules:** Control access to resources.
 - **SSL/TLS Support:** Enables secure HTTPS traffic.
 - **Request Filtering:** Blocks malicious requests.
 - **IP Restrictions:** Restricts access by IP address.
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☒ Logging & Monitoring

- **IIS Logs:** Record HTTP requests, errors, and status codes.
- **Failed Request Tracing:** Helps troubleshoot failed requests.
- **Performance Monitor:** Monitors IIS and app pool performance.
- **Event Viewer:** Tracks system/application events related to IIS.

☒ Common Use Cases

- Hosting static websites.
- Running ASP.NET or PHP-based dynamic applications.

- Serving APIs (RESTful services).
 - Intranet portals in organizations.
 - Reverse proxy for back-end applications (using ARR).
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☒ How to Install IIS on Windows Server

1.
Open **Server Manager**.
2.
Click **Add Roles and Features**.
3.
Select **Web Server (IIS)**.
- 4.

Choose additional features (e.g., ASP.NET, FTP).

5.

Install and configure via **IIS Manager** GUI or **PowerShell**.

☒ Common Tools and Add-ons

Tool	Use
IIS Manager	GUI for configuring and managing IIS.
Web Deploy	Deploy apps and sites from development to production.
Application Request Routing (ARR)	Enables load balancing and reverse proxy.
URL Rewrite	Allows URL manipulation and redirection.

☒ Summary

- IIS is a **Windows-based web server** used to host websites, web applications, and services.
- It supports **application pools, authentication, HTTPS, modules, and logging**.
- IIS is ideal for **ASP.NET apps, enterprise portals, and secure web hosting**.